Changes in meritocratic beliefs among school-age students over time and across cohorts

immediate

Abstract

Besides transmitting academic knowledge, schools are key sites of meritocratic socialization, actively shaping students' understanding of whether success is earned or inherited. Yet the measurement of students' meritocratic beliefs remains underdeveloped: most instruments conflate perceptions of how rewards are allocated with normative preferences for how they should be allocated, and treat merit and privilege as mutually exclusive. This study tests a multidimensional model that distinguishes four dimensions among school-aged students: perceived meritocracy and meritocratic preferences (whether effort and talent are, and should be, rewarded), and perceived privilege and privilege acceptance (whether family wealth and connections shape outcomes, and whether this is seen as legitimate). Using two-wave panel survey data from Chilean students in 8th and 10th grade (Wave 1: N = 846; Wave 2: N = 662), we assess whether these dimensions can be distinguished, compared across school stages, and measured equivalently over time using confirmatory factor analysis and measurement invariance testing. Results support a four-factor structure, showing that endorsement of merit coexists with recognition of privilege rather than displacing it. The scale achieves strict longitudinal invariance within students, enabling valid developmental comparisons, but not across school levels, indicating that older students apply a more critical frame when judging whether effort is rewarded. These findings provide a validated instrument for tracking how meritocratic beliefs develop through schooling. They are relevant for civic education, school climate, and equity-oriented policy, where students' sense of fairness is central: because students can demand merit-based fairness precisely when they perceive its absence, interpreting their beliefs requires distinguishing what students perceive from what they endorse, and merit from privilege, rather than treating belief in meritocracy as a single attitude to reinforce or counter.
Keywords: Meritocracy · Schools · Measurement invariance · Socialization · Chile

1 Introduction

How students come to understand who succeeds and why is one of the central questions in sociology of education (Mijs, 2016; Resh & Sabbagh, 2014). Schools are central sites in this process: they simultaneously promote meritocracy as a cultural ideal, presenting education as the legitimate route to upward mobility and individual effort as the proper basis of social rewards (Van De Werfhorst, 2024), and institutionalize it as a lived experience through grading, tracking, and selection (Darnon, Wiederkehr, et al., 2018; Traini, 2022). Students who believe that effort and talent determine outcomes tend to interpret success and failure in individual terms, with consequences that extend from individual motivation to school climate and beliefs about social inequality (Castillo et al., 2024; Resh, 2018; Weinberg et al., 2021). Yet students also encounter daily evidence that outcomes are shaped by family background, social connections, and structural advantage, creating tensions between meritocratic ideals and lived experience (Bourdieu & Passeron, 1990; Goudeau & Croizet, 2017; Tang et al., 2025). Understanding how students hold these beliefs, whether they see society as meritocratic, whether they think it should be, and whether these judgments extend to privilege, is therefore a question with direct relevance for educational practice and equity in schools.

What makes this particularly complex is that students rarely hold these beliefs in pure form. Meritocratic beliefs carry concrete educational consequences: among teachers, belief in school meritocracy predicts preference for competitive over cooperative classroom practices; among parents, lower demand for equity-oriented policies; and among students, reduced support for interventions aimed at reducing achievement gaps and stronger legitimation of inequality (Batruch et al., 2022; Castillo et al., 2024; Darnon et al., 2023; Darnon, Wiederkehr, et al., 2018; Wiederkehr et al., 2015). However, even students who recognize that family wealth and connections shape outcomes may simultaneously endorse meritocracy as a normative ideal, reflecting what the literature describes as dual consciousness (C. Liu & Wang, 2025; Tang et al., 2025). Whether and how the consequences above differ depending on whether students are expressing perceptions or preferences, and about merit or privilege, is precisely what current research has rarely been able to establish.

Despite this complexity, the field lacks adequate tools to capture these distinctions. Most instruments conflate what students perceive to be true about how rewards are distributed with what they prefer should be true, and treat merit and privilege as opposite ends of a single scale. A student who perceives that only connections and wealth get ahead but strongly believes that effort and talent should determine outcomes holds a fundamentally different orientation from a student who is indifferent to both, yet current instruments would assign them similar scores, misidentifying what students believe and what educational responses should address.

This study addresses this gap with two contributions to educational research. The first is the validation of a multidimensional model of meritocratic beliefs for school-aged populations, building on Castillo et al.'s (2023) framework, which distinguishes between perceptions of how meritocracy operates ("what is") and preferences about how it should ("what ought to be"), and between meritocratic elements (effort, talent) and privilege-based elements (family wealth, personal contacts). Treating these as four distinct but related dimensions, rather than collapsing them into a single score or a bipolar scale (Reynolds & Xian, 2014), allows for belief profiles that unidimensional instruments rule out by design, and provides a stronger empirical basis for studying how schools shape students' orientations toward fairness and inequality. While recent work has begun to treat school meritocracy as multidimensional (Chauvin et al., 2026), no validated instrument yet captures these distinctions among students themselves, a gap this study addresses directly.

The second contribution is to examine whether this instrument functions equivalently across school stages (8th vs. 10th grade) and over time within the same students (2023-2024). Most research on meritocratic beliefs remains cross-sectional, limiting what we can infer about how these orientations develop during adolescence, precisely when reasoning about justice becomes more sophisticated and school-based sorting mechanisms grow more salient (Henry & Saul, 2006; Resh & Sabbagh, 2014). Establishing measurement invariance is a prerequisite for valid developmental comparisons: without it, observed change over time may reflect shifts in how students interpret items rather than genuine change in underlying beliefs (Putnick & Bornstein, 2016; Van De Schoot et al., 2015). Cross-cohort invariance testing serves a complementary purpose: rather than assuming comparability across school stages, we treat the degree of equivalence as an empirical question whose answer is itself theoretically informative about how educational experience reshapes the meaning students attach to meritocratic concepts.

This study draws on panel data from Chile, a setting that is both nationally specific and internationally instructive. Chile's educational system combines high stratification, market-based competition, and explicit meritocratic discourse in school culture (Castillo et al., 2024; Valenzuela et al., 2013), alongside persistent socioeconomic inequality and a deep marketization of social welfare (Ferre, 2023), features shared by many Latin American countries and increasingly present in European and East Asian systems undergoing marketization (Busemeyer, 2014; C. Liu & Wang, 2025). This makes Chile a productive case for studying how students internalize, question, or reconcile meritocratic ideals with the reality of privilege, while the dimensional structure and longitudinal patterns identified here are likely to travel to other highly stratified educational contexts.

The research questions guiding this study are as follows:

(1) Do school-aged students distinguish between meritocratic and privilege-based principles, as well as between perceptions ("what is") and preferences ("what should be")?

(2) Are these dimensions longitudinally stable within students over time, and does their measurement equivalence hold across school stages, or does educational experience reshape the meaning of meritocratic beliefs?

In the following sections, we review the theoretical background on meritocratic beliefs in school settings, present the methodology and results, and discuss the implications of our findings for understanding how meritocratic socialization develops during adolescence and for civic education, school climate, and equity-oriented policy.

2 Theoretical and empirical background

2.1 What is meritocracy and how is it measured?

Meritocracy refers to a distributive system in which individual merit, typically defined as effort and personal ability, is treated as the primary criterion for allocating resources and rewards, rather than social origins or inherited privilege (Bell, 1972; Young, 1958). While Young (1958) coined the term as a dystopian critique, it has since been re-appropriated as a positive ideal of fairness in liberal and market-oriented societies (Mijs, 2021; Van De Werfhorst, 2024). From a sociological standpoint, meritocracy operates both as a cognitive judgment about how inequality works and as a moral lens through which people evaluate whether unequal outcomes are deserved (Castillo et al., 2019; Heuer et al., 2020). Precisely because it frames outcomes as earned, meritocratic ideals can end up reinforcing inequality: "winners" are encouraged to see their position as deserved, while "losers" are pushed toward self-blame rather than structural critique (García-Sierra, 2023; Sandel, 2020). Research in adult populations has documented systematic associations between these beliefs and orientations toward redistribution and educational policy (Castillo et al., 2025; Tejero-Peregrina et al., 2025), but the question of how and when they develop, and whether the same belief structure emerges among school-aged students who are still embedded in the socialization processes that shape them, has received far less attention.

The research agenda on meritocratic beliefs has faced challenges in conceptualizing and measuring meritocracy. Some studies have relied on narrow operationalizations that equate meritocracy with the social valuation of effort (Mijs, 2021; Wiederkehr et al., 2015), often reflecting simplified readings of Young's (1958) formulation (Merit = Effort + Intelligence), whereas others treat it as interchangeable with broader constructs such as system justification (Day & Fiske, 2017; Jost et al., 2004), beliefs about social mobility (Deng & Wang, 2025; McCoy & Major, 2007), or generalized support for equal opportunity (Batruch et al., 2022; Darnon, Wiederkehr, et al., 2018). As a result, conceptually distinct orientations are frequently collapsed into a single indicator, and the distinction between descriptive perceptions of how rewards are allocated and normative preferences about how they should be allocated is often acknowledged but rarely implemented analytically.

A related, more specific limitation concerns how studies account for privilege-based factors such as family wealth, social connections, and other structural advantages. As Castillo et al. (2023) notes, much survey research relies on item batteries that ask respondents to rate the importance of effort, talent, family background, networks, or luck for "getting ahead", and then reduces these into a single score. This becomes especially problematic when researchers construct difference or continuum measures, treating meritocratic and privilege-based explanations as opposite ends of the same scale (e.g., by subtracting "non-merit" from "merit," as in Reynolds & Xian, 2014). Such operationalizations embed a strong zero-sum assumption, namely that endorsing merit necessarily entails rejecting structural or relational advantages, thereby ruling out, by design, the empirical possibility that both types of beliefs coexist. Recent debates, including the exchange between Mijs (2026) and Wiesner & Sachweh (2026), underscore this point by arguing for measures that capture meritocratic beliefs alongside privilege-based ones, rather than as a residual of the relative importance assigned to structural factors. Complementary empirical work reinforces this concern: meritocratic endorsement can remain stable even as recognition of structural constraints increases (C. Liu & Wang, 2025; Tang et al., 2025), and support for merit coexists with recognition of social connections rather than trading off against it (Kwon & Pandian, 2024; Zhu, 2025). Unidimensional indices, and especially subtraction scores, can therefore manufacture artificial neutral positions and blur substantively distinct belief profiles that carry different educational consequences.

In response to these limitations, Castillo et al. (2023) proposed a multidimensional framework that decomposes meritocratic beliefs along two analytically independent axes: preferences versus perceptions, and meritocratic versus privilege-based allocation principles. Preferences capture normative ideals about how rewards should be distributed (e.g., whether effort and ability ought to determine life chances); perceptions capture descriptive evaluations of how society actually works (e.g., whether observed inequalities reflect merit-based processes) (Janmaat, 2013). Meritocratic elements (effort, talent) and privilege-based elements (family wealth, networks) are treated as distinct components rather than as opposite poles of a single continuum. This yields four non-redundant constructs, allowing belief combinations that older measures tend to collapse, such as endorsing meritocracy as an ideal while simultaneously recognizing the weight of family background and connections. The proposal is deliberately minimalist, making it well-suited for large-scale surveys with limited questionnaire space, though this parsimony entails a trade-off: each factor is measured with only two items, which can constrain reliability and content coverage. In what follows, we adopt the terms perceived privilege and privilege acceptance in place of the original "non-meritocratic perceptions/preferences" to avoid defining these constructs by negation and to align with the vocabulary of educational research on structural advantage (Mijs, 2026). Evidence from adult samples using confirmatory factor analysis supports this four-factor structure and shows systematic associations among dimensions: stronger perceived privilege tends to co-occur with stronger meritocratic preferences (Castillo et al., 2023), a pattern that aligns with Zhu's (2025) notion of dual consciousness, here understood as the simultaneous endorsement of meritocracy as a normative ideal alongside descriptive recognition that rewards are shaped by structural advantage.

In school-based research, these measurement challenges have received limited attention, even as the first dedicated instruments have begun to appear. The most direct precedent is Chauvin et al. (2026), who propose a multidimensional scale of school meritocracy and, importantly, model it with a bifactor structure that separates ability and equity dimensions from a general factor. Their work establishes that school meritocracy is not unidimensional, but it also illustrates the gap this study addresses: their scale is developed and validated on teachers rather than students, captures only the perceptual side of meritocracy ("what is"), and does not incorporate privilege-based principles as distinct constructs, so that the perceptions-preferences and merit-privilege distinctions central to our framework fall outside its scope. Castillo et al. (2024) apply the four-factor framework to a sample of Chilean graders and report that students differentiate perceptions from preferences and meritocratic from privilege-based principles; however, that study relies on item-level descriptives and bivariate associations rather than a formal factorial measurement model, meaning the multidimensional structure has not been tested through confirmatory factor analysis in a student population. Beyond this, no study has examined whether these dimensions are stable across age cohorts or over time, which limits understanding of how meritocratic beliefs develop through schooling and whether observed differences across stages reflect substantive change or measurement non-equivalence. This gap is consequential: without an established and tested measurement model, let alone evidence of its equivalence across school levels and waves, findings from different student populations cannot be meaningfully compared, and claims about the development of meritocratic beliefs during adolescence remain on uncertain empirical ground.

2.2 The socialization of meritocracy at school

Schools are central to the socialization of meritocracy because they promote it both culturally and institutionally. On the one hand, they advance meritocratic ideals through narratives of effort, achievement, self-improvement, and education as a legitimate route to mobility and reward (Batruch et al., 2022; Chauvin et al., 2026; Darnon, Wiederkehr, et al., 2018; Wiederkehr et al., 2015). On the other hand, they embed these ideals in routine practices of grading, tracking, selection, awards, and credentialing, which operate as institutional markers of worth, making merit visible and consequential in ways that are immediate and socially meaningful (Resh & Sabbagh, 2014; Traini, 2022). This dual character, at once cultural promotion and institutional enactment, is what makes schools a particularly important arena for understanding how meritocratic beliefs are formed and contested.

At the same time, schools remain settings where privilege-based advantages, including family resources, cultural capital, peer environments, and unequal school quality, are also visible and consequential (Batruch et al., 2022; Goudeau & Croizet, 2017). This duality can give rise to ambivalent or internally differentiated orientations. Students may endorse meritocracy as a normative ideal while simultaneously recognizing structural constraints, reflecting what the literature describes as dual consciousness (C. Liu & Wang, 2025; Tang et al., 2025). This coexistence is not fixed but can shift with educational experience: C. Liu & Wang (2025) argue that education can operate as either legitimation or enlightenment depending on the broader inequality context, and longitudinal evidence suggests that recognition of structural advantage tends to grow as students progress through schooling (Tang et al., 2025). More broadly, meritocratic framing encourages students across social backgrounds to interpret success and failure in individual terms, reinforcing personal responsibility while limiting attention to structural constraints (Darnon, Wiederkehr, et al., 2018). As Lampert (2013) argues, meritocratic schooling can reward a minority while leading many others to internalize failure as a personal deficiency, and even students from disadvantaged backgrounds often adopt meritocratic narratives, suggesting that prolonged exposure to school-based evaluation is a powerful mechanism shaping beliefs about justice and inequality (Henry & Saul, 2006).

These dynamics are not static. Developmental research suggests that by around age 10, children already articulate judgments about social differences, typically emphasizing individual, merit-based explanations while only beginning to incorporate opportunity-based accounts (Imhoff & Brussino, 2025; Sigelman, 2013). As students move through the school system, however, cumulative exposure to grading, tracking, peer comparison, and visible socioeconomic gaps can make the limits of effort as a sufficient explanation for success increasingly apparent. Tang et al. (2025) show that among Chinese college students, those from lower socioeconomic backgrounds initially report lower recognition of privilege-based advantages than their higher-SES peers, but this gap narrows progressively across college years as both groups converge toward stronger acknowledgment of structural advantage. This trajectory suggests something important: it is not simply that older students believe less in meritocracy, but that sustained exposure to the educational system makes structural advantage harder to ignore, generating what has been described as a "reality shock" in which meritocratic discourse increasingly clashes with experienced inequality (Allen, 2016). Adolescence is therefore a particularly consequential period for studying these processes, precisely when abstract reasoning about justice and fairness becomes more sophisticated and school-based sorting mechanisms grow more salient (Henry & Saul, 2006; Resh & Sabbagh, 2014).

The consequences of these beliefs extend well beyond individual attitudes. Students who perceive schools as highly meritocratic, particularly believing that effort and talent are fairly rewarded, are more likely to justify unequal access to social services, perceive class inequalities as less unjust, express weaker support for redistribution, and endorse stronger system-justifying beliefs (Batruch et al., 2022; Castillo et al., 2024; Wiederkehr et al., 2015). The effects are not confined to students: among teachers, belief in school meritocracy predicts preference for competitive over cooperative classroom practices, while among parents it predicts lower demand for policies aimed at reducing socioeconomic inequalities in achievement (Darnon et al., 2023). These patterns suggest that meritocratic beliefs are not merely attitudinal outcomes of schooling but active inputs into the educational environment itself, shaping classroom dynamics, institutional decisions, and the broader normative climate in which students develop.

This body of evidence collectively indicates that meritocratic beliefs in school settings are multidimensional. Whether a student endorses effort and talent as legitimate bases for reward, recognizes that wealth and contacts shape outcomes in practice, prefers a meritocratic ideal, or accepts privilege-based advantage as legitimate are empirically distinct orientations that may develop at different rates, respond to different school experiences, and carry different consequences for academic motivation, civic behavior, and student wellbeing. Measuring them as a single score, or forcing merit and privilege into a zero-sum trade-off, obscures the very complexity that makes these beliefs consequential. A necessary step toward understanding how distinct dimensions of meritocratic belief relate to these outcomes is to establish that they can be measured reliably and equivalently across school stages; something the existing literature has not yet demonstrated. That is the more modest but foundational contribution this study aims to make.

2.3 The Chilean educational context

Chile's educational system has undergone a profound market-oriented transformation since the early 1980s under the military dictatorship (1973-1989), when a comprehensive reform introduced school vouchers, decentralization, and private provision as organizing principles of public education (Valenzuela et al., 2013). The result is a system in which families formally choose among schools competing for enrollment, but in which that competition reproduces and amplifies existing inequalities: school quality is strongly stratified by socioeconomic status, ability tracking begins early, and access to higher-quality institutions depends heavily on academic performance and, in many cases, on family resources and social networks (Bellei, 2013; Moyano Dávila et al., 2025; Otero et al., 2017). Chile consistently ranks among the most educationally stratified countries in the OECD, with large and persistent gaps in learning outcomes across income groups that the expansion of educational access has not resolved (Ferre, 2023; OECD, 2025). Within this structure, meritocratic discourse is actively promoted through institutional design and public policy: effort and individual achievement are framed as the legitimate and sufficient route to social mobility, and recent debates over school admissions reform have reinforced this, with competing governments successfully installing concepts of merit and academic selection at the center of public debates about educational fairness, displacing alternative framings centered on inclusion and equity (Cabalin et al., 2023).

Within this context, students, teachers, families, and adults who have navigated the system all tend to combine meritocratic ideals with recognition of structural advantage, though the form this takes varies by social position and school type. Among secondary students, awareness of educational inequality is widespread, yet meritocratic principles are simultaneously invoked to evaluate and naturalize one's own position within it (Neut Aguayo et al., 2025). Among teachers and parents, merit is defined less through academic outcomes than through attitude and visible effort, with failure attributed to individual will rather than structural conditions, a pattern that varies by school type and was accentuated after the pandemic (Moyano Dávila et al., 2025). Even among upwardly mobile professionals, success is rarely attributed to individual merit alone: effort is understood as a collective family project, and luck, networks, and institutional support are recognized as equally decisive (Fercovic, 2022). Together, these findings suggest that the coexistence of meritocratic and privilege-based accounts is a durable feature of Chilean culture, making the school system the primary arena in which these orientations are first encountered and gradually consolidated.

Beyond its national specificity, Chile's case speaks to broader patterns in comparative educational research. The combination of market-based competition, ability tracking, explicit meritocratic discourse, and persistent socioeconomic stratification that characterizes the Chilean system is not unique to Latin America: similar configurations have emerged in European countries that have pursued school choice and accountability reforms (Busemeyer, 2014; Gingrich, 2011), and in China, where high-stakes selection mechanisms produce strong meritocratic narratives alongside intense competition and visible class-based advantage (C. Liu & Wang, 2025; Tang et al., 2025). This makes Chile not a peripheral case but a particularly clear instance of a more general phenomenon: the coexistence of meritocratic ideology and structural inequality in formally open educational markets. How meritocratic beliefs are structured, how stable they are across school stages, and whether they can be measured equivalently across student populations are questions that any research program on meritocratic socialization must confront, regardless of national context. Findings on how to measure these beliefs reliably in a highly stratified system can therefore inform similar efforts elsewhere and contribute to building a comparative basis for understanding how schooling shapes students' orientations toward merit, privilege, and educational opportunity.

2.4 Hypotheses

Building on Castillo et al.'s (2023) framework and extending it to a longitudinal, multi-cohort design, this study tests three hypotheses:

H_1 **(Dimensional differentiation)**. Students' meritocratic beliefs are best represented by four distinct factors: perceived meritocracy, perceived privilege, meritocratic preferences, and privilege acceptance.

H_2 **(Between-cohort measurement invariance)**. The four-factor model is invariant across age cohorts.

H_3 **(Longitudinal measurement invariance)**. The four-factor model is invariant across waves within students.

3 Method

3.1 Participants

This study uses secondary data from the 2023 and 2024 waves of the Panel Survey on Education and Meritocracy (EDUMER), which examines school-age students' beliefs, attitudes, and behaviors regarding meritocracy, inequality, and citizenship. The questionnaire was administered to sixth-grade and first-year secondary students in nine schools in Chile's Metropolitan and Valparaíso regions. Although the sample was non-probabilistic and lacked quota controls, a minimum target of 900 students was set to ensure adequate statistical power. After data processing and listwise deletion, the analytical sample included 846 students in Wave 1 (386 girls, 421 boys, 39 identifying as other; M_{age} = 13.4, SD_{age} = 1.6) and 662 in Wave 2 (303 girls, 338 boys, 21 identifying as other; M_{age} = 14.4, SD_{age} = 1.6), yielding a retention rate of 78.3% (attrition = 21.7%). The main sources of attrition were school transfers between academic years and high absenteeism in several schools.

3.2 Procedure

Data were collected by a professional research firm using a Computer-Assisted Web Interviewing (CAWI) approach, based on online questionnaires. All participants received informed consent, which was reviewed and approved by a parent or legal guardian. To enable longitudinal linkage while protecting confidentiality, the survey firm generated anonymous unique identifiers for each student, and no directly identifying information was included in the analytic files.

3.3 Measures

3.3.1 Scale of perceptions and preferences of meritocracy

Meritocratic and privilege-based perceptions and preferences were measured using the same items proposed in the original scale (Castillo et al., 2023). Each of the four dimensions of the scale is measured by two items. The dimensions are:

- Meritocratic perceptions: the extent to which effort and ability are rewarded in Chile,
- Privilege perceptions: the extent to which success is perceived as linked to connections and family wealth,
- Meritocratic preferences: agreement with that those who work harder or are more talented should be better rewarded,
- Privilege acceptance: agreement that it is acceptable for individuals with better connections or wealthy parents to achieve greater success (see Table 1).

Each item is rated on a four-point Likert scale ranging from "strongly disagree" (1) to "strongly agree" (4), therefore higher scores indicate stronger endorsement of the corresponding perception or preference.

Table 1: Items for perceptions and preferences for meritocracy scale

Components	Dimensions	Item (English)	Item original (Spanish)
Perception	Meritocratic	In Chile people are rewarded for their efforts	En Chile las personas son recompensadas por sus esfuerzos
		In Chile people are rewarded for their intelligence and ability	En Chile las personas son recompensadas por su inteligencia y habilidad
	Privilege	In Chile those with wealthy parents do much better in life	En Chile a quienes tienen padres ricos les va mucho mejor en la vida
		In Chile those with good contacts do much better in life	En Chile a quienes tienen buenos contactos les va mejor en la vida
Preference	Meritocratic	Those who work harder should reap greater rewards than those who work less hard	Quienes más se esfuerzan deberían obtener mayores recompensas que quienes se esfuerzan menos
		Those with more talent should reap greater rewards than those with less talent	Quienes poseen más talento deberían obtener mayores recompensas que quienes poseen menos talento
	Privilege	It is good that those who have rich parents do better in life	Está bien que quienes tengan padres ricos les vaya mejor en la vida
		It is good that those who have good contacts do better in life	Está bien que quienes tengan buenos contactos les vaya mejor en la vida

3.3.2 Cohort level

To distinguish students' academic level across waves for conditional and multi-group invariance analyses, we created a cohort indicator capturing whether the respondent belonged to the primary (sixth grade) or secondary (first-year secondary) cohort at the time of each survey wave. Descriptive statistics by cohort and wave are reported in Table 2.

Table 2: Descriptive statistics for cohort level

Wave	Cohort level	N	Age		Gender		
			Mean	SD	Man	Women	Others
Wave 1	Primary	406	11.85	0.56	195	191	20
	Secondary	440	14.79	0.80	226	195	19
	Total	846	13.38	1.63	421	386	39
Wave 2	Primary	319	12.83	0.55	161	145	13
	Secondary	343	15.76	0.75	177	158	8
	Total	662	14.35	1.61	338	303	21

3.4 Analytical strategy

3.4.1 Measurement model and estimation

To evaluate the underlying structure of the scale, we employed Confirmatory Factor Analysis (CFA) based on a measurement model with four latent factors, using Diagonally Weighted Least Squares with robust correction (WLSMV) estimation. This estimator is particularly suitable for ordinal data, such as four-point Likert-type scales, as it avoids the bias associated with treating categorical data as continuous (Kline, 2023).

Model fit was assessed following the guidelines of Brown (2015), using several indices: the Comparative Fit Index (CFI) and the Tucker-Lewis Index (TLI), with acceptable values above 0.95; the Root Mean Square Error of Approximation (RMSEA), with values below 0.06 indicating good fit; and the Chi-square statistic (acceptable fit indicated by $p > 0.05$ and a Chi-square/df ratio < 3).

3.4.2 Measurement invariance

A key contribution of this study is its assessment of measurement stability through factorial invariance testing (Putnick & Bornstein, 2016). We implemented two complementary strategies: longitudinal invariance across two panel waves and cross-group invariance across school levels (primary vs. secondary students). Because the items are ordinal, we followed recent recommendations for ordered categorical indicators, specifically the longitudinal approach of Y. Liu et al. (2017) and the cross-group procedure of Svetina et al. (2020). In the longitudinal analyses, we estimated the conventional hierarchy of configural, metric, scalar, and strict models, progressively constraining factor loadings, thresholds/intercepts, and residual variances over time. In the cross-group analyses, however, we followed the more appropriate three-step sequence for ordinal indicators proposed by Svetina et al. (2020), estimating a configural model, then a threshold-invariant model, and finally a model jointly constraining thresholds and factor loadings across groups.

In addition to the chi-square difference test, we assessed invariance using changes in approximate fit indices. Following Chen

(2007), we treated ΔCFI as the primary criterion and ΔRMSEA as supplementary, indicating noninvariance when $\Delta CFI \geq -0.010$ and $\Delta RMSEA \geq 0.015$.

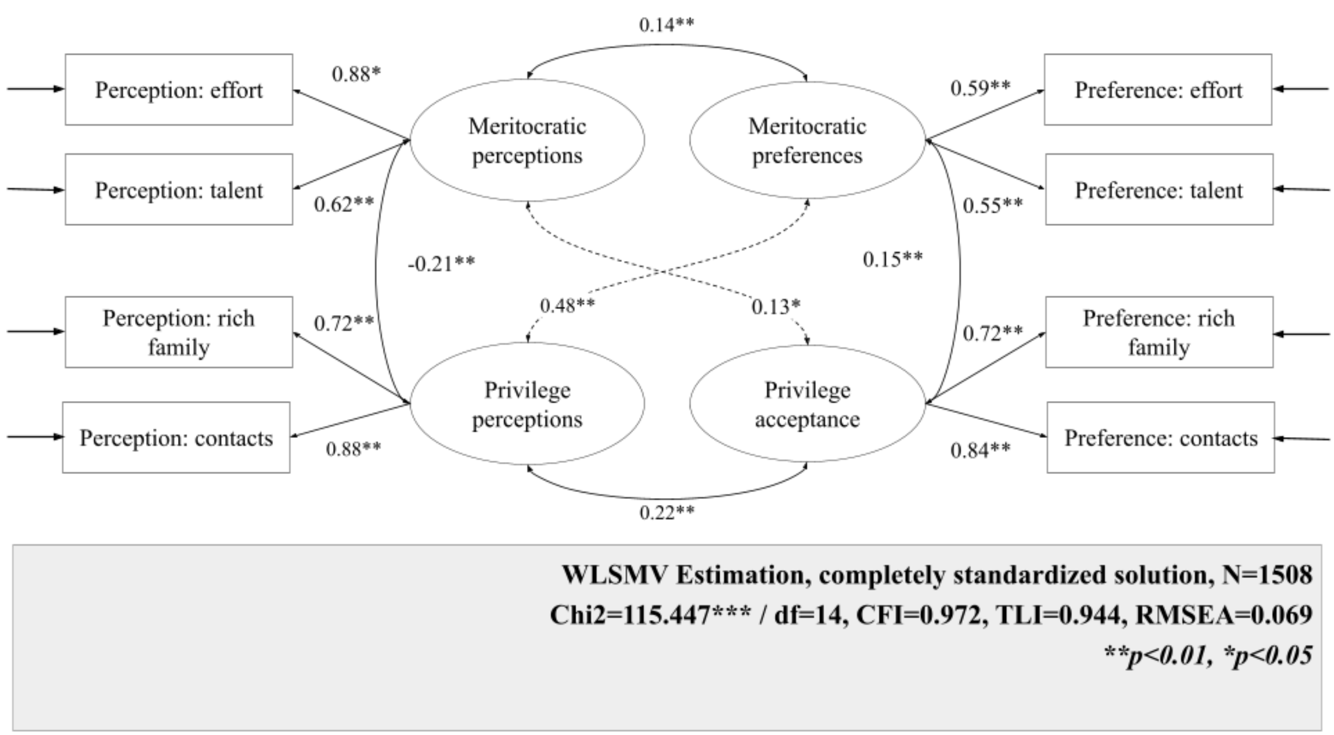
All analyses were performed using the *lavaan* package in R version 4.5.2. The hypotheses of this research were pre-registered on the Open Science Framework (OSF).

4 Results

4.1 Measurement model

Figure 1 presents the standardized factor loadings and the overall fit indices of the general measurement model estimated on the pooled analytical sample across cohorts and waves. The model shows a reasonable, though not fully satisfactory, fit to the data (CFI = 0.97; TLI = 0.94; RMSEA = 0.07). Although the CFI falls within commonly accepted ranges, the RMSEA and TLI remain below conventional thresholds, indicating that fit is acceptable in some respects but not uniformly strong across indices. Overall, these results suggest that the proposed four-factor structure captures the main covariance pattern in the combined sample reasonably well [1].

Figure 1: Multigroup model diagram



All standardized factor loadings are statistically significant and generally moderate to strong, though their magnitude varies across dimensions. For meritocratic perceptions, both indicators load positively on the latent factor, with effort showing a particularly strong loading ($\beta = 0.88$) and talent a moderately strong loading ($\beta = 0.62$). Meritocratic preferences exhibit more

[1] Factor loadings and fit indices for the separate analytical samples (i.e., per wave and cohort level) are available in the Supplementary Material.

even, but somewhat weaker, loadings for effort (β = 0.59) and talent (β = 0.55). For perceived privilege, both indicators load strongly, especially contacts (β = 0.88), followed by coming from a rich family (β = 0.72). Privilege acceptance also shows strong and relatively balanced loadings, with contacts (β = 0.84) and rich family (β = 0.72), indicating that both indicators are similarly representative of this latent dimension.

The latent correlations reveal a differentiated structure among the four dimensions. Meritocratic perceptions and meritocratic preferences are positively correlated (r = 0.14), consistent with some alignment between what students perceive is rewarded and what they believe should be rewarded. Meritocratic perceptions and privilege perceptions are negatively correlated (r = −0.21), suggesting that students who perceive greater meritocracy in society tend to view a weaker role for connections and family wealth. At the same time, privilege perceptions are positively related to meritocratic preferences (r = 0.48, strongest correlation), suggesting that perceiving society as rewarding contacts and wealth may coincide with stronger endorsement of meritocratic ideals. Finally, meritocratic preferences and privilege acceptance are modestly positively correlated (r = 0.15), and privilege perceptions and preferences are also positively associated (r = 0.22), indicating that recognizing privilege in practice is linked to a greater acceptance of their legitimacy, albeit to a moderate extent.

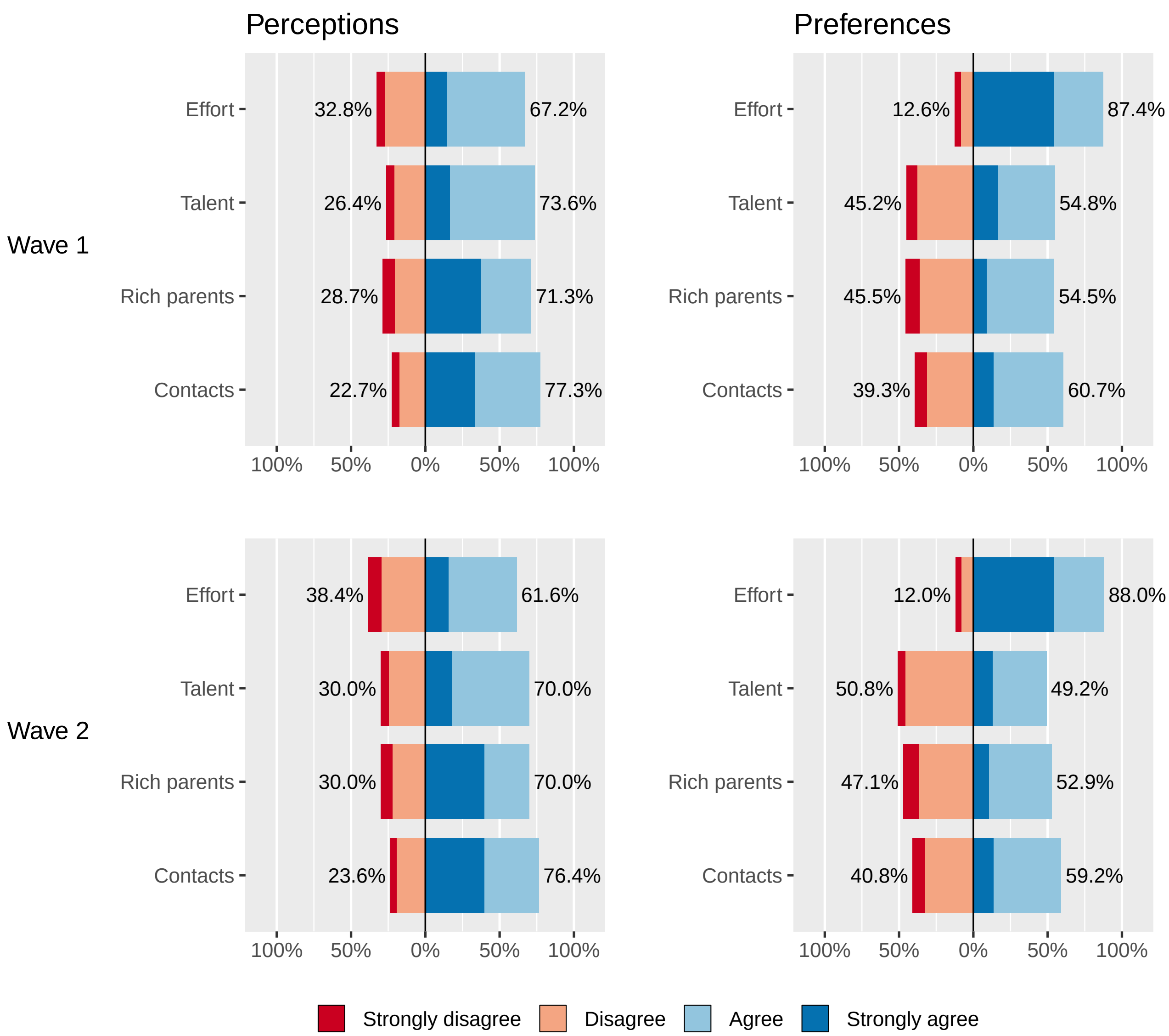
4.2 Invariance analysis

4.2.1 Longitudinal invariance

Figure 2 displays the response distributions for the meritocracy scale, distinguishing perceptions (Panel A) from preferences (Panel B) across waves. Across both waves, a clear and stable gap emerges between how students perceive social rewards to operate and how they think they should operate. In the perception items, agreement is consistently high for contacts, parental wealth, and talent, whereas effort is also widely endorsed but to a lesser extent, indicating that students recognize both meritocratic and privilege determinants of success and view talent as more influential than effort. In the preference items, by contrast, effort is overwhelmingly endorsed as the legitimate basis for rewards, far exceeding its perceived role in practice. Talent occupies a more ambivalent normative position, parental wealth remains evenly divided, and contacts are more often accepted than rejected, suggesting that privilege criteria are not uniformly seen as illegitimate. Wave 2 closely replicates these patterns, with only modest shifts. Overall, the descriptive results reveal a persistent paradox: strong endorsement of effort-centered meritocratic ideals alongside skepticism about their realization in practice.

Table 3 reports the results of the longitudinal invariance tests. Overall, the scale achieved the strongest level of invariance (strict invariance) across the two waves. The strict model showed good fit, $\chi^2(84)$ = 130.17, CFI = 0.991, RMSEA = 0.031 (90% CI: 0.020–0.040). Moreover, compared with the strong (scalar) model, the deterioration in fit was negligible: $\Delta\chi^2(4)$ = 1.635, ΔCFI = 0.000, and ΔRMSEA = -0.002. These differences fall well within commonly used cutoffs for invariance evaluation (Chen, 2007), indicating that constraining residual variances in addition to loadings and thresholds/intercepts is tenable. Substantively, this implies that the item–factor relations are stable over time (metric invariance) and that, for a given level of the latent trait, respondents are expected to endorse the same response categories across waves (scalar invariance), supporting the comparability of scores and latent parameters within students over time.

Figure 2: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by wave



Source: own elaboration based on EDUMER Panel Survey (N Wave 1 = 846; N Wave 2 = 662)

Table 3: Longitudinal invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Configural	117.7 (68)	0.991	0.035 (0.024-0.046)	0 (0)	0	0.000	Reference
Weak	122.51 (72)	0.990	0.034 (0.024-0.045)	4.809 (4)	0	-0.001	Accept
Strong	128.53 (80)	0.991	0.032 (0.021-0.042)	6.02 (8)	0	-0.002	Accept
Strict	130.17 (84)	0.991	0.031 (0.02-0.04)	1.635 (4)	0	-0.002	Accept

As an additional diagnostic, we used univariate score tests to identify potential sources of localized misfit at each step of the invariance hierarchy. The tests revealed no meaningful violations of the imposed equality constraints, suggesting that model fit was not driven by a small subset of parameters. Global score tests and the smallest p-values at each step are reported in the Supplementary Material.

4.2.2 Cohort invariance

Figure 3 shows response distributions by cohort, separating perceptions and preferences. In primary education, effort and talent are perceived as similarly rewarded, while contacts and family wealth are also seen as advantageous, though with more disagreement. In secondary education, perceptions become less meritocratic: agreement that effort is rewarded declines, whereas contacts and family wealth are seen as more strongly shaping success. Preferences are more stable across cohorts. In both groups, effort is the most strongly endorsed legitimate basis for reward, while talent, family wealth, and especially contacts remain more ambivalent. Notably, contacts are more often accepted than rejected in both cohorts. Overall, the cohort comparison reveals a widening gap between an effort-centered meritocratic ideal and the perceived importance of privilege advantages in practice.

Table 4: Cohort (multigroup) invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Configural	24.95 (26)	0.993	0.036 (0.009-0.057)	0 (0)	0.000	0.000	Reference
Tresholds	57.56 (42)	0.980	0.049 (0.033-0.064)	43.875 (16) ***	-0.013	0.013	Reject
Tresholds + Loadings	72.35 (46)	0.973	0.053 (0.039-0.067)	17.075 (4) **	-0.006	0.005	Reject

Figure 3: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by cohort

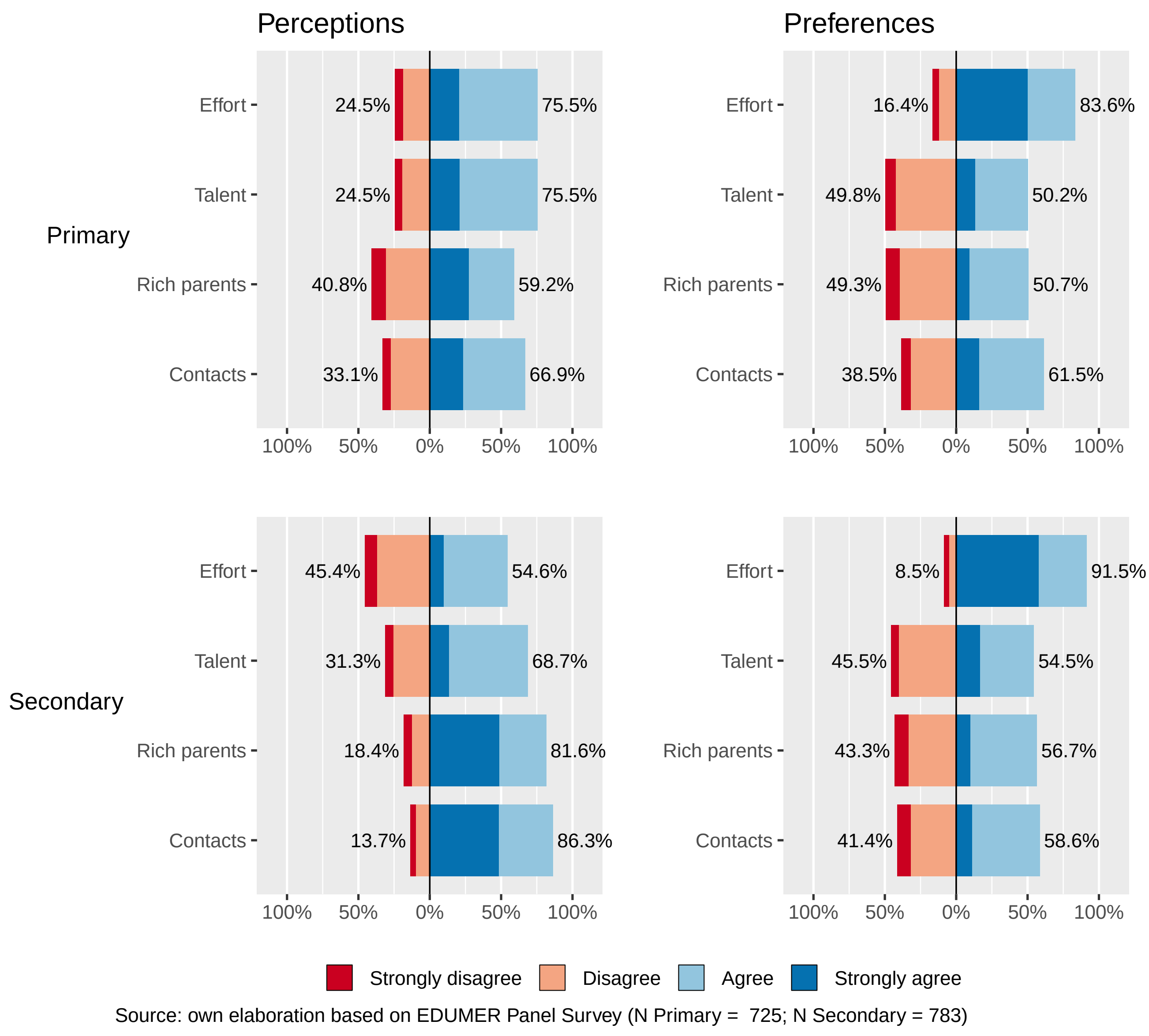


Table 4 summarizes the cross-cohort invariance tests. The scale achieves configural invariance between primary and secondary students, but not threshold invariance, and therefore does not meet the prerequisites for stronger forms of invariance (Svetina et al., 2020; Wu & Estabrook, 2016). Constraining thresholds worsened model fit (ΔCFI = $-.013$; ΔRMSEA = .013), exceeding recommended cutoffs (Chen, 2007). This indicates that response categories do not map equivalently onto the latent constructs across cohorts, limiting direct comparisons of latent means or other latent differences.

To locate the source of cross-cohort non-invariance, we inspected localized misfit in the threshold-invariant model. The largest violations concerned the perceived-effort item, especially the second threshold ($\chi^2 = 10.12, p = .001$) and, to a lesser extent, the first ($\chi^2 = 7.55, p = .006$), suggesting that older students may apply a more critical standard when judging whether effort is rewarded in society.

Despite the lack of cohort invariance, re-estimating the longitudinal invariance models with cohort as a covariate predicting each latent factor did not materially change model fit or conclusions, suggesting that invariance over time remains tenable. These robustness checks are reported in the Supplementary Material.

5 Discussion

The first hypothesis proposed that students' meritocratic beliefs would be best represented by four distinct factors: perceived meritocracy, perceived privilege, meritocratic preferences, and privilege acceptance. The confirmatory factor analyses support this expectation: the scale performs well in the school-aged population, with satisfactory model fit, strong factor loadings, and meaningful correlations among factors. Consistent with evidence from adult populations (Castillo et al., 2023), these results indicate that even at an early stage of school socialization, students distinguish between perceptions and preferences, and between meritocratic and privilege-based principles. This four-factor structure also resonates with qualitative evidence from Chilean schools, where students across socioeconomic groups simultaneously invoke meritocratic ideals and recognize structural advantage (Neut Aguayo et al., 2025), and where merit is often defined by attitude and visible effort rather than by academic outcomes (Moyano Dávila et al., 2025). This last finding is relevant for interpreting the privilege acceptance dimension: in contexts where merit is primarily understood as behavioral disposition, accepting that privilege-based advantages lead to better outcomes may not be experienced as a contradiction but as a separate normative judgment about a different domain, a reading that, while going beyond what Moyano Dávila et al. (2025) directly establish, is consistent with their evidence. This suggests that the coexistence of these orientations reflects how students actually navigate competing narratives about fairness and reward, rather than an artifact of survey design.

The empirical separability of these dimensions has substantive implications for how meritocratic beliefs are understood and measured (Kwon & Pandian, 2024; Mijs, 2026; Zhu, 2025). Perceived and preferred meritocracy are already distinguishable among school-aged students, consistent with the distinction between "what is" and "what should be" and with earlier attempts to operationalize it (Castillo et al., 2023; Duru-Bellat & Tenret, 2012). This matters because the same normative endorsement of merit can carry different meanings depending on perceived reality: high perceived meritocracy is more compatible with affirming existing arrangements, whereas low perceived meritocracy can reflect dissatisfaction and a demand for greater fairness. At the same time, meritocratic and privilege-based principles coexist rather than forming opposite poles of a single continuum, directly challenging zero-sum measurement strategies that presuppose endorsing merit entails rejecting privilege (Reynolds & Xian, 2014; Wiesner & Sachweh, 2026). The correlational pattern in our data reinforces this: perceived privilege is positively associated with stronger meritocratic preferences, suggesting a normative-reactive response to perceived unfairness (Castillo et al., 2023). This is consistent with Zhu's (2025) argument that meritocratic and structural explanations accumulate rather than displace one another, and with accounts in which merit endorsement may remain stable as recognition of structural constraints increases (C. Liu & Wang, 2025; Tang et al., 2025). It is also worth noting how this finding relates to typological approaches: Kwon & Pandian (2024) document three distinct attitudinal clusters in Europe, showing that coexistence takes different forms across individuals. The present study complements that work by showing how the same coexistence can be represented within a single coherent measurement model that supports longitudinal inference, rather than requiring person-centered classification. This has concrete implications for how educators and policy designers interpret students' orientations toward fairness. A student who strongly endorses merit as an ideal while recognizing that privilege shapes outcomes is not disengaged or fatalistic: they hold a normative demand for fairness that the system is not meeting. Conflating this profile with one of simple disbelief in effort leads to misdiagnosis, whether in classroom practice or in policy design. Distinguishing which dimension is at stake, endorsement of merit, recognition of privilege, or acceptance of it as legitimate, matters for how schools interpret students' sense of fairness and respond to it through civic education, classroom climate, and efforts to build trust in the fairness of school itself.

These results also have implications for how the existing literature on school meritocracy should be interpreted. A substantial body of research has linked stronger meritocratic beliefs among students to inequality legitimation, weaker support for redistribution, and greater acceptance of unequal access to educational resources (Batruch et al., 2022; Castillo et al., 2024; Wiederkehr et al., 2015), while among teachers and parents, similar beliefs predict competitive classroom practices and lower demand for equity-oriented policies (Darnon et al., 2023). A particularly relevant finding is that perceived, but not preferred, school meritocracy predicts lower support for egalitarian and inclusive educational interventions (Darnon, Smeding, et al., 2018), and that belief in school meritocracy is associated with greater acceptance of social-class achievement gaps (Darnon, Wiederkehr, et al., 2018). However, most of this research relies on unidimensional or undifferentiated measures, which means it cannot establish whether these effects are specific to particular dimensions or reflect a more general meritocratic orientation. Our findings suggest that these documented effects are likely dimension-specific: perceived meritocracy is not equivalent to preferred meritocracy, and neither carries the same consequences as perceived privilege or privilege acceptance. The contrast with Chauvin et al. (2026) is instructive here, as theirs is the closest existing instrument: by distinguishing ability and equity within a bifactor model, they establish that school meritocracy is multidimensional, but because their measure is confined to teachers and to perceived meritocracy, it cannot speak to whether students separate what they perceive from what they prefer, or merit from privilege. What the present study adds is a four-factor structure validated in students and shown to be longitudinally invariant, which provides the measurement foundation needed to examine whether distinct dimensions of meritocratic belief carry distinct consequences for the educational environment, from students' sense of justice in school to civic attitudes and the broader climate of fairness, a question the existing literature has not yet been able to address with adequate precision.

The second hypothesis stated that the measurement model would be stable over time, and the results support this expectation. Achieving strict longitudinal invariance indicates that the scale functions equivalently across waves, measuring the same latent constructs at each time point. This is crucial in developmental research because without invariance, apparent change may reflect shifts in how students interpret items or use response categories rather than genuine change in underlying beliefs (Y. Liu et al., 2017; Putnick & Bornstein, 2016; Van De Schoot et al., 2015). With strict invariance in place, observed differences over time can be interpreted as substantive within-student change rather than measurement artifact, strengthening longitudinal inference and supporting models such as latent mean comparisons or growth curve analyses (Putnick & Bornstein, 2016). Substantively, the results suggest that meritocratic belief structures are relatively stable over a one-year window in early-to-mid adolescence, which has implications for when educational interventions targeting these beliefs are likely to produce measurable change. This longitudinal invariance also provides the measurement foundation for subsequent studies to examine how distinct dimensions of meritocratic belief relate to outcomes such as academic motivation, civic attitudes, and school well-being. In this regard, Liu & Wang's (2025) distinction between education operating as legitimation or as enlightenment, depending on the broader inequality context, points to a productive direction: whether perceived privilege leads to stronger meritocratic preferences or to disengagement from institutional narratives may depend on the degree to which the educational system is experienced as fair, a question that requires dimensionally validated instruments to examine adequately.

The third hypothesis proposed invariance across educational levels. The results did not support full metric or scalar equivalence: non-invariance is concentrated in perceived meritocracy, specifically in how students at different school stages interpret whether effort is rewarded in society. Rather than treating this as a measurement failure, we interpret it as a substantive theoretical finding. Cross-cohort non-invariance indicates that primary and secondary students do not merely differ in their level of perceived meritocracy: they attach qualitatively different meanings to the concept of effort being rewarded. This is consistent with developmental research showing that adolescence is precisely the period when abstract reasoning about justice and fairness becomes more sophisticated and when school-based sorting mechanisms, including grades, tracking, and peer comparison, grow increasingly salient (Henry & Saul, 2006; Resh & Sabbagh, 2014), providing students with a richer and more critical experiential basis for evaluating meritocratic claims. A plausible mechanism is that cumulative exposure to these processes produces a progressive reality shock (Allen, 2016; Tang et al., 2025), such that older students evaluate the same item through a more critical and experience-laden lens. If the SES-differentiated convergence documented by Tang et al. (2025) operates earlier, during secondary school in Chile's highly stratified system, the non-invariance concentrated in the perceived-effort item may partly reflect students across social positions increasingly confronting the limits of effort as a sufficient explanation for outcomes. In Chile's highly stratified system, where the gap between meritocratic discourse and structural reality is particularly visible, students from lower socioeconomic backgrounds have been shown to experience this gap as a moral grievance (Neut Aguayo et al., 2025), which is consistent with the idea that the construct of perceived effort becomes more contested, and therefore less stable across students, as schooling advances. For researchers, this is a call to move beyond assuming that meritocracy scales travel unchanged across school levels: future work should prioritize stage-sensitive item formulations and partial invariance approaches that can accommodate meaning shifts without sacrificing cross-level comparability (Putnick & Bornstein, 2016; Van De Schoot et al., 2012). For practitioners, including teachers and those designing civic education and school-climate initiatives, the finding indicates that the same meritocratic message may be processed through different interpretive frames depending on students' accumulated school experience, and that efforts to foster a shared sense of fairness in school need to be calibrated to developmental stage rather than applied uniformly across grade levels.

Taken together, the longitudinal and cross-cohort findings are not contradictory but jointly informative. Longitudinal invariance was established within the same students over a one-year window: within that period, the scale measures the same constructs in the same way, and observed change can be interpreted as substantive. Cross-cohort non-invariance reflects a different comparison, two groups separated by approximately two to three years of accumulated schooling, and points to a longer developmental process through which cumulative school experience progressively reshapes how students interpret whether effort is rewarded. The instrument is therefore well-suited for longitudinal inference within the same school stage, while cross-stage comparisons require caution and, ideally, stage-sensitive measurement approaches.

6 Conclusion

Schools socialize students into beliefs about meritocracy, promoting effort and talent as the legitimate basis of reward while simultaneously exposing students to the weight of family resources and social connections in shaping outcomes (Goudeau & Croizet, 2017; Resh & Sabbagh, 2014; Traini, 2022). Students navigate this tension by holding both orientations at once, endorsing merit as an ideal while recognizing that privilege structures actual outcomes, a configuration the literature describes as dual consciousness (C. Liu & Wang, 2025; Tang et al., 2025). Capturing this complexity requires instruments that distinguish how students perceive meritocracy to work from how they prefer it should work, and that treat merit and privilege as potentially coexisting rather than mutually exclusive, a standard most available measures do not meet. Extending Castillo et al.'s (2023) multidimensional framework to adolescents, this study tests whether students distinguish merit from privilege and perceptions ("what is") from preferences ("what should be"), using panel survey data from two Chilean cohorts (8th and 10th graders) across two waves (2023-2024), and whether these dimensions are stable over time and comparable across school stages. The contribution is a modest but foundational one: to establish whether meritocratic beliefs can be measured reliably and equivalently in school-aged populations, as a precondition for studying their developmental trajectory and educational consequences.

The results support a four-factor structure, perceived meritocracy, perceived privilege, meritocratic preferences, and privilege acceptance, showing that Chilean adolescents differentiate perceptions from preferences and meritocratic from privilege-based principles, and that merit endorsement can coexist with recognition of structural advantage. Consistent with this, perceived privilege is positively associated with stronger meritocratic preferences, suggesting that recognizing unfair advantages can intensify demands that rewards should follow merit, a normative-reactive pattern that unidimensional instruments would obscure (Castillo et al., 2023; Mijs, 2026; Wiesner & Sachweh, 2026). The measurement structure is longitudinally invariant within students, providing a reliable basis for interpreting over-time change as substantive rather than artifactual. At the same time, cross-cohort non-invariance is itself a substantive finding rather than merely a technical caveat: comparisons across educational stages require caution because the perception that "effort is rewarded" does not carry the same meaning for primary and secondary students. This measurement heterogeneity is theoretically consistent with developmental accounts in which cumulative schooling experience progressively reshapes how students interpret meritocratic signals (Allen, 2016; Tang et al., 2025), and it calls for stage-sensitive item formulations in future work rather than treating the current scale as directly comparable across levels. In doing so, the instrument moves beyond the most recent dedicated alternative, the teacher-focused, perception-only scale of Chauvin et al. (2026), toward a measure that captures how students themselves distinguish merit from privilege and description from preference.

More broadly, these findings sharpen how the literature should interpret meritocracy effects in educational contexts. By distinguishing dimensions that unidimensional measures collapse, the four-factor model shows that part of what is often read as a decline in belief in meritocracy may instead reflect growing recognition of structural constraints rather than weaker merit endorsement (C. Liu & Wang, 2025; Reynolds & Xian, 2014; Tang et al., 2025; Zhu, 2025). It also offers a distinct route to the coexistence of contrasting beliefs: where typological approaches document that coexistence by sorting individuals into attitudinal clusters (Kwon & Pandian, 2024; Zhu, 2025), the present study represents it within a single measurement model that supports longitudinal inference, rather than requiring person-centered classification. The combination of strict longitudinal invariance and cross-level non-invariance establishes a key boundary condition: within-student comparisons over time are warranted, whereas comparisons across school stages must account for the shifting meaning of perceived meritocracy.

Although this study draws on a single country, its contributions carry implications beyond the Chilean case. Three findings are likely to travel across educational systems. First, the conceptual distinction between perceived and preferred meritocracy, and between meritocratic and privilege-based principles, reflects general cognitive and normative capacities that develop during adolescence regardless of national context; the four-factor structure should therefore replicate in other highly stratified systems. Second, the coexistence of merit endorsement and recognition of privilege, the core pattern underlying dual consciousness, has been documented in China (C. Liu & Wang, 2025; Tang et al., 2025), the United States and Finland (Zhu, 2025), and Europe (Kwon & Pandian, 2024), suggesting it is not a Chilean peculiarity. Third, the logic of longitudinal invariance testing as a prerequisite for developmental inference is methodologically general and applies to any panel study of meritocratic beliefs. By contrast, the specific pattern of cross-cohort non-invariance, concentrated in perceived effort, may be more system-dependent: it likely reflects the particular intensity and timing of school-based sorting in Chile’s tracked and marketized system. Future research in countries with different tracking regimes and stratification profiles will help clarify which aspects of measurement equivalence are universal and which are conditioned by institutional features.

The findings carry distinct implications for three audiences. For researchers, the results underscore the importance of treating meritocratic beliefs as multidimensional and of testing measurement invariance before making developmental or cross-group comparisons, since otherwise observed differences may reflect differential item interpretation rather than genuine attitudinal divergence, particularly across socioeconomic or generational groups. For practitioners, including teachers, school counselors, and those who design civic and citizenship education, the coexistence of merit endorsement and recognition of privilege suggests that students are not passive recipients of meritocratic ideals but actively navigate tensions between what schools tell them about fairness and what they observe in practice. This has direct relevance for civic education and for school climate: rather than simply reinforcing effort-based narratives or, conversely, dismantling them, educators can engage students’ existing dual awareness as a starting point for discussing how merit and privilege coexist, helping cultivate a more critical and grounded sense of justice. Because the meanings students attach to effort and reward shift across school stages, such efforts need to be calibrated to students’ developmental moment rather than applied uniformly. For policymakers concerned with equity, the finding that perceived privilege intensifies meritocratic preferences carries a specific message: students who recognize that rewards do not follow merit do not abandon the ideal of fairness but demand it more strongly. Educational systems that maintain a strong meritocratic discourse while visibly reproducing inequality risk widening the gap between what they promise and what students experience, with potential consequences for students’ sense of justice in school, their trust in educational institutions, and their broader civic engagement.

This study has limitations that point to clear directions for future research. The scale is deliberately minimalist, with two items per factor, which constrains content coverage and reliability and means that claims about subtle within-dimension differences should be treated with caution. Future work should expand and refine the item pool, adding at least one additional indicator per factor and testing whether richer item sets alter the invariance conclusions reported here. With only two waves and two school levels, additional waves and broader age coverage are needed to better trace developmental trajectories and identify when shifts in meaning in perceived meritocracy occur. Cross-level non-invariance calls for differential item functioning analyses and stage-sensitive item formulations to support comparability across educational stages. Most importantly, future research should examine how perceptions and preferences of meritocratic and privilege-based principles relate to the outcomes for which they are most likely to matter in school: students' sense of justice and fairness, civic attitudes and engagement, school climate, and wellbeing, alongside academic motivation. By providing an empirically supported and longitudinally invariant multidimensional measure for adolescents, this study offers a more reliable foundation for studying how school-based meritocratic socialization shapes students' interpretations of fairness and inequality.

7 Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude (Anthropic) to assist with language editing, restructuring, and condensing of the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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